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Bringing a Highly Automated Land Drilling Rig into One of the World's Oldest Oilfields to Prepare for the Low Carbon Future: The Why and How

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Abstract

Automated mass production has historically driven enormous gains in global wealth and well-being by reducing unit costs through economies of scale, much of it built on the foundation of fossil fuels. Today, as the world pivots toward a low-carbon future, this industrial model must evolve to support the next energy revolution. Wells are essential to this transition, enabling not only traditional hydrocarbon production but also geothermal energy, hydrogen storage, and carbon capture and storage (CCS). A key challenge is whether the advantages of automated mass production can be extended further into drilling, enabling the delivery of low-margin green energy projects through low-cost, high-quality wells.

To address this, the Operator has pioneered the introduction of a highly automated land drilling rig, effectively a well manufacturing system, into one of the world's oldest oil-producing regions, Romania. The rig features a manless rig floor during routine operations such as drilling and tripping, craneless rig-up and rig-down using remotely operated pin connections, and fully trailer-mounted equipment. Together, these innovations promise a step change in cost per well, operational speed, and consistency, enabling fewer rigs to drill more wells while reducing carbon footprint and environmental impact.

Analysis of real-time drilling and rig move data indicates significant efficiency gains in brownfield development if the rig's performance potential is consistently realized. The Operator, in collaboration with the contractor, is using these projects as a learning platform to prepare for the factory drilling demands of the energy transition.

The rig deployment and startup were executed through close cooperation between the Operator, contractor, and rig manufacturer, with multidisciplinary teams working toward a shared vision. Achieving alignment, from overarching goals to individual tasks, proved essential in avoiding delays and misunderstandings. Importantly, managing the cultural shift toward a new way of working, where decades-old habits and expectations had to be reshaped, made motivation, communication, and change management the most critical success factors throughout the project.

This paper will describe the technical rationale, operational setup, and, most importantly, the communication and change management strategies that led to the successful spud of the first well.

Introduction

Romania's Oil Heritage

Romania has one of the oldest oil industries in the world. Crude oil production was officially recorded as early as 1857, and the first industrial refinery began operation in Ploiesti in 1856. Today, Romania continues to produce hydrocarbons from mature brownfields, some of which have been in continuous operation for over a century. These fields have aging infrastructure, complex reservoirs and narrow economic margins, which create operational and safety challenges. These conditions make automation increasingly relevant as a way to maintain safe and efficient drilling while supporting environmental goals.

Industry Drivers for Automation (HSSE, Efficiency, and Sustainability)

Modern drilling operations face high expectations for safety, consistent performance, cost control and sustainability. Automation helps meet these expectations by reducing the need for people to work in hazardous areas and by delivering repeatable, precise execution of routine tasks. This allows people to focus on planning and supervision rather than manual handling, which improves both safety and efficiency. Automated systems also reduce variability between operations, leading to more predictable well delivery and supporting energy-efficient rig designs that can lower the environmental footprint of drilling.

OMV's DigitUP Strategy and Decision to Adopt Automated Drilling

OMV launched its E&P digitalization initiative called DigitUP in 2018 with the ambition to become a digital frontrunner in the upstream oil and gas sector. Its aim was to bring digital technologies across the entire upstream value chain, enabling safer, more efficient and more sustainable operations.

As part of DigitUP, a bold and visionary concept for well delivery was established: "plan the well in a day, drill it with robots and control it with people". This was envisioned as a future state where planning becomes rapid and data-driven, drilling tasks are automated, and human expertise remains essential for supervision and decision making. It was intentionally aspirational, serving as a rallying call to stretch current capabilities and guide forward-thinking and innovation.

The RigUP project was conceived as the flagship initiative to deliver on the "drill it with robots" part of the vision. It focused on deploying a fully automated land rig, demonstrating how automation can augment human strengths and reduce physical exposure. RigUP became the tangible project that connected DigitUP's aspirational vision with real-world execution in operational settings.

Scope of the paper

This paper explains how the automated land rig was implemented, covering market research, tender design, organizational change and field deployment. Its goal is to share insight into the implementation process, highlight lessons learned and provide recommendations for similar projects. The paper does not focus on technical performance benchmarking, but on how the automated rig was integrated into existing operations and what cultural and organizational changes were needed for success.

Project Strategy and Methodology

Concept and Drivers

The automation project was an outcome of a well delivery strategy to drive efficiency with latest digital and robotic technology. The project idea was born in 2019 to look for a robotic rig as the physical body capable of being controlled by artificial intelligence in expectation of a possibility to drill autonomously in the future. Market research for the rig started in 2020-2021 and the tender was launched and concluded in 2022.

Several drivers influenced the decision:

- First, and of course most important, was the need to improve safety by removing people from high-risk tasks and zones. This was to be achieved both during drilling operations and during rig moves. By introducing automation, the requirement for personnel to perform repetitive manual tasks in hazardous areas was eliminated. Streamlined processes reduced the frequency and duration of human presence on the rig floor and around suspended loads, ensuring that people were exposed to danger zones only when strictly necessary. The principle was clear, people should spend as little time as possible in high-risk areas, focusing instead on supervision and decision-making roles while machines handled routine execution.
- A vision for factory drilling capability, supporting faster and more consistent well delivery. This meant adopting automation to standardize repetitive tasks such as tripping, tubular handling and rig-up and rig-down. Considering the pre-existing conditions, this vision translated into the potential to increase annual well delivery from approximately 11 wells per rig to as many as 19, providing a step change in development efficiency.
- The goal to reduce well delivery costs while maintaining or improving quality. Automation and performance improvements were expected to reduce the time and resources required per well, leading directly to lower unit costs. This was particularly important in Romanian brownfields, where narrow economic margins make cost efficiency a decisive factor in sustaining drilling activity. Unlike the productivity focus of factory drilling, the cost driver emphasized making each well more affordable without compromising safety or technical integrity.
- The broader energy transition context, where efficient well construction is required for future geothermal, hydrogen storage and carbon capture applications. Strategically, cheaper wells would open up these new business opportunities where drilling costs play a major role of up to 80% of the total project.

Together, these elements formed a strong case for piloting a fully automated land rig in Romania, laying the foundation for a forward-looking project aimed at redefining land rig operations.

Project Strategy

When considering where to deploy such a rig, OMV Petrom emerged as the best candidate because of its high drilling activity and diverse portfolio of wells. By analyzing the company's drilling history, the project team identified the well types most frequently drilled and used this information to define the required rig capacity and features. This ensured the automated rig would be optimized for the majority of OMV Petrom's drilling schedule rather than for specialized or rare well designs.

The project followed a well-defined, phased strategy designed to align technical insight with regulatory compliance and business risk mitigation. The initial step involved conducting comprehensive market research to assess the maturity, availability, and suitability of automated drilling technologies, particularly for use in Romanian brownfield environments. Insights from this analysis informed the development of a detailed public tender scope that accurately reflected operational needs, market conditions, and innovation potential. The next phase consisted of executing all procedural stages required by Romania's public procurement law.

This deliberate sequencing ensured that by the time the project reached the stage for final investment decision, the team had full visibility into actual technical capabilities and financial proposals submitted in the tender. Crucially, no binding contractual commitments had been made at that point, allowing the operator to assess risks and benefits with minimal exposure. As a result, the business case for deployment was grounded in real commercial data rather than assumptions, significantly increasing the investment team's confidence and enabling a better-informed approval of the automated rig concept.

The initial market research focused on technology rather than contracting models. OMV engaged directly with drilling rig manufacturers to understand what automation technologies were available and what level of integration was technically feasible for land rigs. This work helped define the desired features of an automated rig and served as input for preparing the scope of work for a public procurement process in Romania.

To attract the right candidates for this serious investment in automation, the tender was structured with several special features:

- A commitment of 1000 operating days over a period of 5 years of operations, providing the operational stability needed to justify investment in a highly automated rig.
- A bonus/malus performance contract with KPIs selected to ensure the benefits of automation would be fully realized during operations, encouraging continuous improvement and optimal rig utilization.
- A positioning of the project as a unique opportunity to acquire and operate a next-generation automated rig with strong support from the operator during deployment and early operations.

This combination of technical clarity, long-term work commitment and performance-based incentives successfully drew interest from contractors not only in Romania but also across Europe, highlighting the attractiveness of the project and its potential to advance automation in land drilling.

Partnership Model

The winning bid came from Foraj Sonde S.A. Videle (FSV), which proposed deploying Huisman's HM150 automated land rig. This solution met the project's automation requirements and aligned with OMV and OMV Petrom long-term vision for well delivery.

The partnership model on the project was structured as follows:

- OMV E&P Headquarters served as the initiative owner, providing strategic direction, securing approvals and coordinating the implementation effort across multiple regions and disciplines.
- OMV Petrom acted as the operator and end-user, integrating the automated rig into its mature brownfield operations and providing the operational environment for its deployment.
- Foraj Sonde S.A. Videle became the rig owner and operator, responsible for daily drilling operations, rig moves, logistics and meeting defined performance targets.
- Huisman supplied the HM150 automated land rig itself, including the rig design, automation systems and technical support during commissioning and early operations.

This setup created a complex working relationship because of the way the contracts linked the parties to one another, adding significant intricacy that extended beyond technical delivery. OMV Petrom, as the operator, contracted directly with FSV for drilling services, while FSV held the separate contract with Huisman for rig supply, commissioning and ongoing technical support.

Managing this dynamic required more than conventional project oversight. Regular trilateral workshops were organized to bridge contractual boundaries and establish joint priorities, while escalation pathways were agreed upfront to resolve issues without jeopardizing rig delivery or deployment. Informal diplomacy played as much of a role as formal governance, ensuring that trust was built across teams despite differing contractual incentives. Ultimately, the shared ambition to pioneer automation in Romania's mature fields acted as the unifying element, allowing the consortium to transform a potentially fragmented setup into a collaborative model. This experience demonstrated that introducing disruptive technology is not only a matter of engineering, but equally one of aligning complex stakeholder relationships under a single operational vision.

The Drilling Rig

Rig Concept

The HM150 is a next-generation automated land drilling rig designed for safe, efficient, and environmentally conscious operations. The "150" indicates a hook load capacity of 150 short tons (136 metric tons), providing sufficient capability for the medium-depth wells that dominate OMV Petrom's brownfield portfolio [3].

Unlike conventional rigs, the HM150 is built around automation. Routine tasks such as tubular handling, tripping, and connection make-up are performed by automated systems controlled from a centralized cabin, eliminating the need for personnel on the rig floor and reducing exposure to high-risk operations.

Key Automation Features

The HM150 integrates multiple automated systems through a SCADA-based control environment (table 1).

Table 1—Automation features of HM150 Rig

Feature	Description	Benefits
Manless Rig Floor	Automated arms, elevators, and an integrated iron roughneck handle tubulars, removing people from hazardous zones during tripping and casing operations.	Eliminates human presence in hazardous zones, improves safety.
Automated Tripping	Achieves slip-to-slip connection times of 80-95 seconds with predictable and repeatable pipe handling.	Faster, consistent operations with reduced variability.
Casing Handling & Casing Drilling	Runs casing in-house without third-party crews; supports casing drilling for wellbore stability or cost reduction.	Lowers operational costs, improves wellbore stability, reduces reliance on third parties.
Automated Mud Bucket	Integrated system with sensors and hydraulic actuators that automatically engages during connections/disconnections, capturing and redirecting drilling fluid.	Reduces personnel exposure to fluids, improves rig floor cleanliness, minimizes slip hazards.
Top Drive Automation	Modified top drive trolley automates positioning, torque management, and connection make-up.	Improves connection integrity, operational consistency, and efficiency.
Centralized Operator Cabin	Climate-controlled cabin monitors and controls all automated functions.	Enhances situational awareness, ergonomics, and supervisory efficiency.

Mobility and Footprint

The HM150 is a fully trailerized system designed for rapid mobilization (table 2).

Table 2—Mobilization features of HM150 Rig

Feature	Description	Benefits
Trailer Count	Eight dedicated trailers carry mast, drill floor with integrated pipe handler, BOP handling system, rig support module, mud treatment system, pump module, power generation, and VFD house.	Simplifies logistics, ensures modularity, enables faster mobilization.
Craneless Rig-Up/Rig-Down	Built-in hydraulic systems and pin connections remove the need for heavy-lift cranes.	Reduces rig move complexity, increases safety, lowers mobilization costs.
Compact Pad Layout	Deployed footprint of ~44.4 m × 40.5 m allows operation on smaller well pads.	Minimizes earthworks, reduces environmental impact, enables access to constrained sites.
Rig Move Time	Designed for 6-8 hour rig moves, covering rig-down, transport, and rig-up.	Rapid deployment, improved operational efficiency, reduced downtime.

Innovative High-Performance Features

Beyond its automation and trailerized design, the HM150 incorporates several innovative features that enhance operational performance and rig site safety:

- **Dual Drawworks System** - The HM150 employs twin drawworks drums, allowing independent control of the traveling block and auxiliary hoisting functions. This configuration provides faster top drive hoisting and lowering speeds, improves load distribution, and offers redundancy in case one system requires maintenance. Importantly, the dual-drum design eliminates the need for traditional slip-and-cut operations, reducing nonproductive time and further improving rig floor safety. As a result, overall tripping efficiency is increased, and complex well programs—such as those requiring frequent tool changes or heavy casing strings—can be handled with greater operational flexibility and smoother load control.
- **LED Lighting Systems** - Energy-efficient integrated LED lights are permanently installed on the top of the mast eliminating the need for temporary lighting towers that are typically erected and wired separately at conventional rig sites. This integrated lighting solution provides superior visibility, consumes less power compared to halogen systems, and reduces the time and complexity associated with setting up lighting infrastructure during rig mobilization. Enhanced illumination supports safe operations during night shifts and adverse weather conditions, improving situational awareness across the worksite.
- **BOP Slide Installation** - The HM150 features a dedicated hydraulic skidding system for installing and removing the blowout preventer (BOP) stack. This system allows the BOP to be slid into position with precision alignment capabilities, eliminating the need for heavy external lifting equipment such as cranes. By integrating this functionality into the substructure, the time required for BOP installation or removal is significantly reduced, while the exposure of personnel to high-risk suspended loads is minimized. The result is a safer, faster, and more controlled critical path activity during rig-up and maintenance operations.

These features collectively contribute to improved operational efficiency, enhanced rig safety, and reduced environmental and logistical footprint, reinforcing the HM150's position as a high-performance automated drilling platform.

Organizational and Cultural Change

Introducing a fully automated rig into conventional operations was not only a technical challenge but also an organizational one, requiring people and processes to adapt in parallel with the new technology. To manage this transition, a dedicated team was formed within OMV Petrom to lead the move toward automated operations. This group, composed of a leader, engineers and drilling supervisors, was deliberately handpicked to combine operational expertise with openness to innovation. The concept was to create an "A-Team" that could act as the nucleus of change, bridging the gap between field crews, project management and the technology provider. Team members received targeted training on automation principles, safety management in automated environments and new logistics workflows, equipping them to champion the rig concept and to provide direct support during deployment. Their role was not only technical but also cultural, demonstrating commitment, setting the tone for adoption and serving as early adopters who could transfer knowledge to the wider workforce.

The deployment of the HM150 automated rig triggered a profound transformation not only in operational workflows but also in organizational culture and technical practices. This transition was supported by a structured change management approach that included a three-phase rig upgrade program, covering both mechanical and control system modifications. Live demonstrations, such as the Casing While Drilling operation attended by OMV Petrom's executive leadership, played a key role in building trust and validating

the rig's capabilities. As automation proved reliable, personnel were progressively withdrawn from the rig floor, marking a shift toward remote and hands-off operations [3].

Initial resistance from field personnel was addressed through clear demonstrations of safety improvements. The integration of automated mud bucket systems and hands-free pipe handling significantly reduced exposure to hazardous tasks. Ergonomics were improved through centralized control, and performance benchmarking confirmed the benefits, with Non-Productive Time (NPT) reduced to below 1% in some cases.

Workforce training was a cornerstone of the transition. Crews were trained to operate the SCADA interface and interpret real-time performance indicators, such as Rate of Penetration and Mechanical Specific Energy. New logistics workflows were introduced, including the use of pre-assembled bottom hole assemblies and casing components, which streamlined operations and reduced manual handling. The adoption of automated and semi-automated tripping and casing running procedures further reinforced the shift toward digital rig control.

The project also required close technical collaboration between OMV Petrom, Foraj Sonde S.A. Videle, and Huisman. Key upgrades included the disassembly and structural modification of the pipe handler gripper, the extension of the top drive's parallel arm to improve alignment and reduce friction, and the redesign of the degasser unit to support crane-less rig moves. These enhancements were critical to enabling safe and efficient operations under the new automated model.

As a result of these changes, supervisory roles evolved toward data-driven decision-making, while field crews adapted to remote diagnostics and digital oversight. The introduction of new logistics models, such as the use of Combilift multi-directional lifting system and pre-torqued BHAs, reduced physical labor and increased operational efficiency. Together, these organizational and cultural shifts laid the foundation for scalable, high-performance drilling.

Field Deployment Experience

In 2024, OMV Petrom launched a transformative drilling campaign featuring the deployment of the HM150 rig (Fig. 1), designed by Huisman. During the initial operational phase, several wells were successfully drilled across Romania, including both appraisal and development wells. Compared to prior deployments, the new campaign demonstrated significant improvements in automation, mobility, and digital integration. Field results demonstrated a 30% average increase in net ROP with the automated rig, leading to time savings of up to 5 days over 10,356 meters drilled across five wells. Comparative analyses confirmed the rig's superior efficiency across various lithologies and well profiles (figures 2 and 3). [1]



Figure 1—Automated Drilling Rig HM150

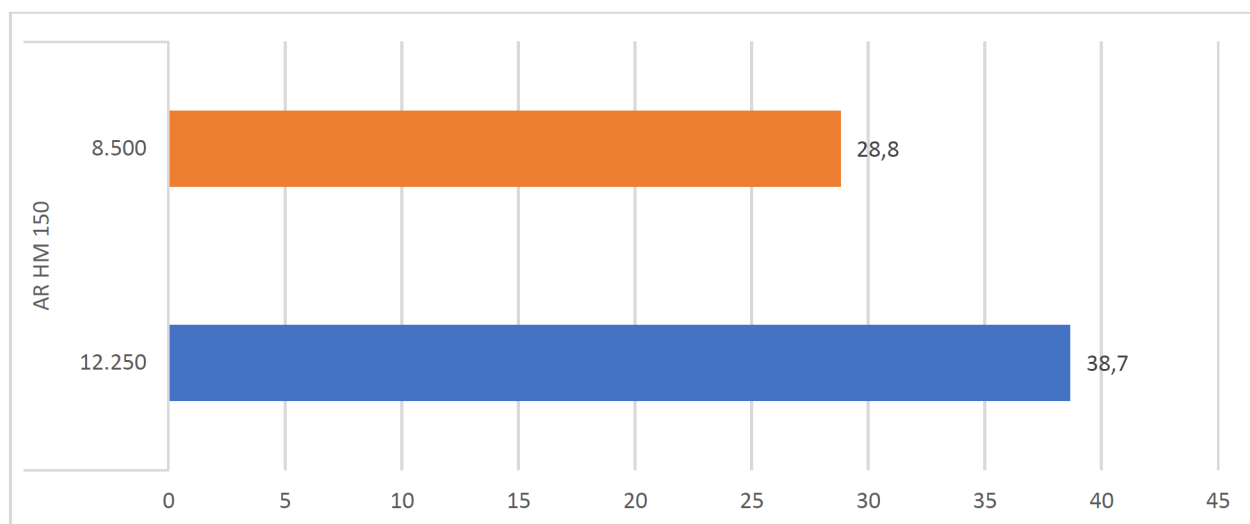


Figure 2—Average - Net ROP per Stand, Avg (m/h) - Automatic Rig

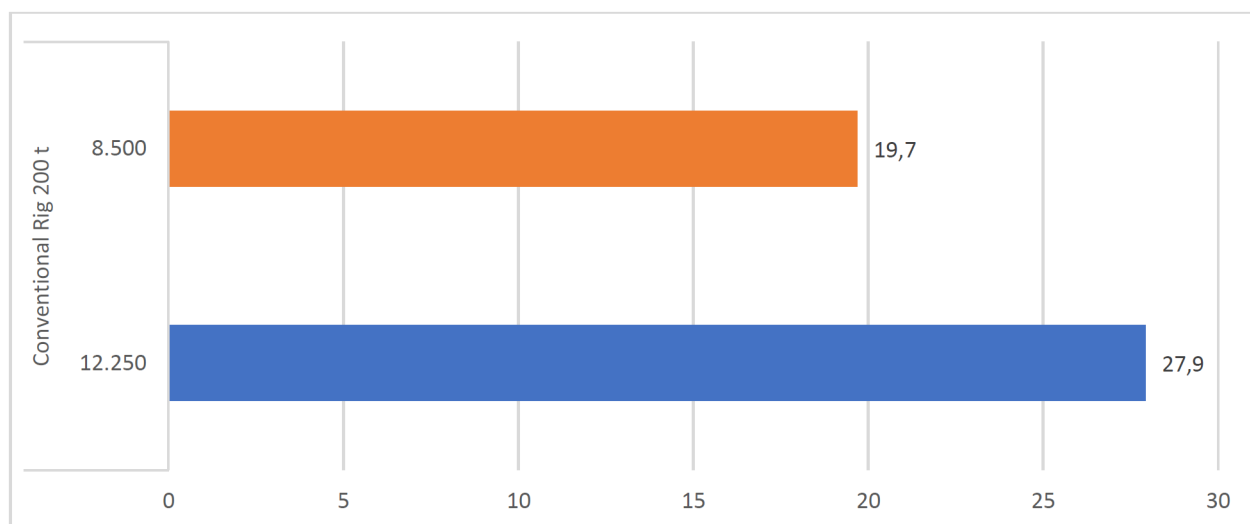


Figure 3—Average - Net ROP per Stand, Avg (m/h) - Conventional Rig

The trailer-mounted and fully electric architecture of the HM150 rig necessitated re-engineering site logistics to accommodate fast rig moves and modular setup. Sites were prepared to support craneless rig-up and rig-down, eliminating heavy auxiliary equipment and reducing mobilization time. Additionally, the switch from diesel-powered generators to a centralized electric system required tailored adjustments to local infrastructure for energy delivery and data connectivity [4].

Commissioning efforts focused on integrating the SCADA-based control system and calibrating the automated driller's real-time response to downhole conditions. In mature brownfield settings, infrastructure limitations were overcome by installing fiber-optic communication networks and connecting the rig's OPC-UA server to OMV Petrom's cloud-based digital ecosystem. This setup enabled seamless real-time data acquisition and dynamic control during early operational stages.

Rig crews underwent an extensive training program that included digital simulations, scenario-based exercises, and data analytics instruction. Emphasis was placed on transitioning crew members from manual tasks to roles focused on supervisory oversight and interpretation of real-time KPIs, such as Rate of Penetration (ROP), Mechanical Specific Energy (MSE), and connection time benchmarks. These efforts

ensured alignment with the rig's automated processes and reduced human intervention in hazardous operations.

Deploying the HM150 automated rig in aging oilfields required compatibility between legacy systems and advanced digital platforms. Real-time data flow between the coherent well planning solution, cloud-based drilling reporting system, and field devices was established, ensuring continuous alignment from planning through execution and post-well review. Infrastructure challenges, such as data routing and electric power availability, were addressed to support the rig's operational demands.

Key challenges included navigating lithological variability and complex well profiles, which required dynamic drilling parameter adjustments. This was addressed through automation embedded in the rig's control system, enabling automatic mode switching (rotary, sliding, geosteering) based on formation feedback. Another major hurdle was the cultural shift among field personnel, some of whom were initially skeptical of fully automated operations. This was managed through live performance demonstrations, transparent communication of efficiency gains, and inclusive training that positioned crews as active participants in digital decision-making loops.

Lessons Learned and Recommendations

While much attention in drilling projects is given to technical specifications and performance, the HM150 experience showed that success also depends on less tangible factors. The triangular relationship between operator, drilling contractor, and rig manufacturer introduced contractual and operational complexity. Although such dynamics may seem secondary compared to engineering challenges, they proved to be of paramount importance. Success required more than technical coordination, it depended on diplomacy, joint decision-making and shared ownership of issues across parties linked by separate contracts rather than a single agreement. Establishing common goals and creating forums for regular alignment proved essential to overcoming these barriers.

The deployment of the HM150 rig revealed several key success factors that contributed to the project's positive outcomes. Technical upgrades, including the automated mud bucket, top drive modifications, and degasser redesign, directly enhanced safety and operational efficiency. The introduction of new logistics workflows, such as pre-assembled bottom hole assemblies and casing components, significantly reduced handling time on site. Remote operations, enabled by the SCADA system and supported by pad redesign and infrastructure improvements, allowed for safer and more streamlined execution.

However, the project also faced notable challenges. Mechanical modifications, such as the disassembly and welding of the gripper unit, required precise coordination and technical expertise. Training on hydraulic and automated systems proved essential but demanded substantial resources and time. Additionally, environmental and visibility constraints, including winterization needs and the impact of mud tank roofs on shaker visibility, required adaptive planning and flexible solutions.

Based on these experiences, several recommendations can be made for other operators considering automated rigs. It is crucial to plan mechanical upgrades in phases and to test each modification under real operational conditions. Predictive maintenance and SCADA-based diagnostics should be leveraged to minimize downtime and ensure system reliability. Modular logistics solutions, such as multi-directional lifting systems and run-ready casing, can further streamline operations and reduce personnel exposure to high-risk tasks.

Looking ahead, the HM150's craneless, trailerized design and automated workflows make it an ideal candidate for scaling to other applications. Its adaptability and efficiency make it suitable for geothermal wells with tight cost margins, CCS (Carbon Capture and Storage) projects that require repeatable execution, and remote brownfields where minimizing personnel exposure is critical.

Conclusion

The introduction of the HM150 automated rig into one of the world's oldest oilfields represents more than just a technical achievement; it demonstrates how vision, collaboration and perseverance can reshape established ways of working. By combining bold ambition with disciplined execution, the project has shown that automation can make drilling safer, more efficient and more sustainable without losing the human creativity and decision-making that remain at the heart of the industry. The lessons learned in Romania provide not only a roadmap for future automated rig deployments but also a message of confidence that the industry can adapt, transform and thrive as it enters a new energy era.

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Special recognition is given to the project team members who contributed their time, expertise and innovative thinking to make the introduction of automation in a mature onshore environment possible. Their efforts demonstrate the value of collaboration and shared vision in achieving transformational change in drilling operations.

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